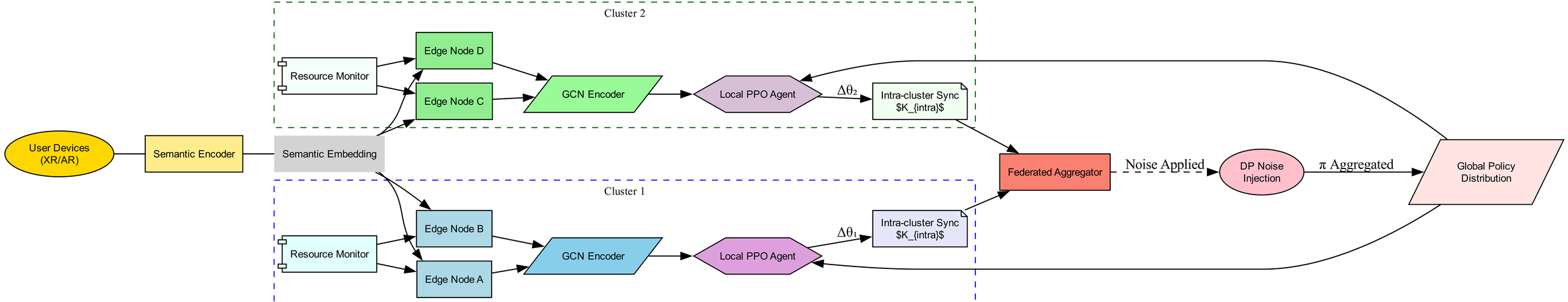
Cluster 2
Edge Node D
Resource Monitor
Edge Node C
GCN Encoder
Local PPO Agent
$\Delta\theta_2$
Intra-cluster Sync
K_{intra}
User Devices
(XR/AR)
Semantic Encoder
Semantic Embedding
Federated Aggregator
Noise Applied
DP Noise
Injection
π Aggregated
Global Policy
Distribution
Cluster 1
Edge Node B
Resource Monitor
Edge Node A
GCN Encoder
Local PPO Agent
$\Delta\theta_1$
Intra-cluster Sync
K_{intra}